Causal Forecast-Conditioned BS-RoFormer for Ocean Sound Separation

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***Abstract*—We evaluate FC-BSR with one fixed main setting: a causal GRU, FiLM, a BS-RoFormer separator, and 5-s windows. Each supervised example is (H_t, N_t, s_t, x_t): twelve immediately preceding 5-s background features whose annotated whale-call intervals are masked, an annotation-clean real 5-s ocean segment, a synthetic marine event, and their exact mixture. The available recordings provide 146.4 h of call-free background, with 37.3 h eligible for training. The 12-step GRU reduces held-out forecast MAE from 2.111 to 1.926 dB (8.8%) against persistence. On 4,096 held-out clean-background mixtures, FC-BSR is 0.134 dB below the unconditioned separator (day-block 95% CI $[-0.320, 0.125]$), with no detectable average improvement. A separate 13-family study compares the other conditioning modes as ablations. On untouched real calls, where no isolated stem exists, a frozen event classifier finds no clear semantic advantage over the matched unconditioned separator. We therefore support fully supervised separation on annotated-clean backgrounds while leaving the value of the forecast prior on unknown real events unresolved.**

***Index Terms*—ocean acoustics, source separation, soundscape forecasting, BS-RoFormer, weak-reference evaluation**

I. Introduction

Ocean backgrounds vary with shipping, weather, biology, and season [1]. A mixture-only separator must infer both an unexpected event and this changing background from the same observation. We instead ask whether the preceding soundscape can supply a causal prior. Our main data tuple is

$$\begin{aligned} &(H_t, N_t, s_t, x_t), \quad x_t = N_t + s_t, \\ &H_t \xrightarrow{\text{GRU}} \widehat{S}_t, \qquad x_t \xrightarrow[\widehat{S}_t]{\text{FiLM+BSR}} (\hat{s}_t, \hat{N}_t), \end{aligned} \tag{1}$$

where H_t excludes annotated call intervals and N_t is a real current no-call segment. Training has exact stems, while the separator never receives N_t or s_t as input.

Time-domain separators [2], [3], attention models [4], [5], and band-split systems [6], [7] provide strong mixture-only backbones. Hybrid Transformers and MixIT broaden the architecture and supervision regimes [8], [9], but do not test a forecast of the upcoming background. FiLM, cross-attention, and cross-modal separation [10]–[12] motivate three interfaces. Forecasting baselines range from recurrent to linear and patch-based models [13]–[16].

Our contributions are: (1) FC-BSR,[1] with a single pre-registered main setting (GRU+FiLM+BS-RoFormer, 5 s); (2) a reproducible annotation-aware construction of 146.4 h of current call-free ocean background and event-masked 5-s histories; (3) a 13-family marine-aware event generator; and (4) a frozen-classifier evaluation on untouched real calls. TOL-domain, conditioning-mode, time-scale, and unannotated-background strength studies are reported as ablations rather than as the primary evidence.

[1]Code and reproduction instructions: https://github.com/D-eval/ocean_noise_pred_splite.

II. Method

A. Forecast and conditional separator

The main history $H_t \in \mathbb{R}^{12\times 30}$ contains 30-band log-power features computed on twelve preceding 5-s windows; annotated calls and their 2-s guards are excluded. Features are normalized using training data only. A residual GRU predicts

$$z_t = \mathrm{LN}(\mathrm{GRU}(H_t)), \quad \widehat{S}_t = S_{t-1} + g_\psi(z_t). \tag{2}$$

Residual initialization represents persistence exactly; a learned predictor is retained only if validation MAE improves upon persistence [17].

The main FC-BSR maps the mixture STFT to band tokens X_k and applies FiLM before the axial Transformer:

$$\widetilde{X}_k = (1 + \tanh \gamma_k) \odot X_k + \beta_k. \tag{3}$$

The separator predicts an event stem; the ocean stem is the mixture residual. Training combines waveform L_1 and multi-resolution complex-STFT losses. Concatenation and cross-attention are alternative conditioning ablations, not the deployed FC-BSR configuration.

B. Marine-aware source distribution

The generator contains 13 families: harmonic, linear chirp, inharmonic, impulse train, frequency-modulated tone, band noise, whale upsweep, downsweep, arch, S-curve, pulse packet, vessel tonal, and vessel cavitation. Whale proxies use one to three localized events with variable duration, curvature, partials, taper, and flutter. Vessel-tonal and narrow-band cavitation components draw a 5–20 Hz low-frequency-oscillator amplitude envelope with probability 0.8; their broadband component is not modulated. Training SNR is uniform on $[-18, 12]$ dB. This distribution is intended to cover contour and modulation patterns, not to claim species identity.

We quantify spectral displacement with directional $D_{\mathrm{KL}}(p\|q)$ and finite symmetric Jensen–Shannon (JS) divergence [19]; spectral entropy and source–ocean overlap are reported separately. Larger KL/JS expands coverage but is not treated as semantic diversity.

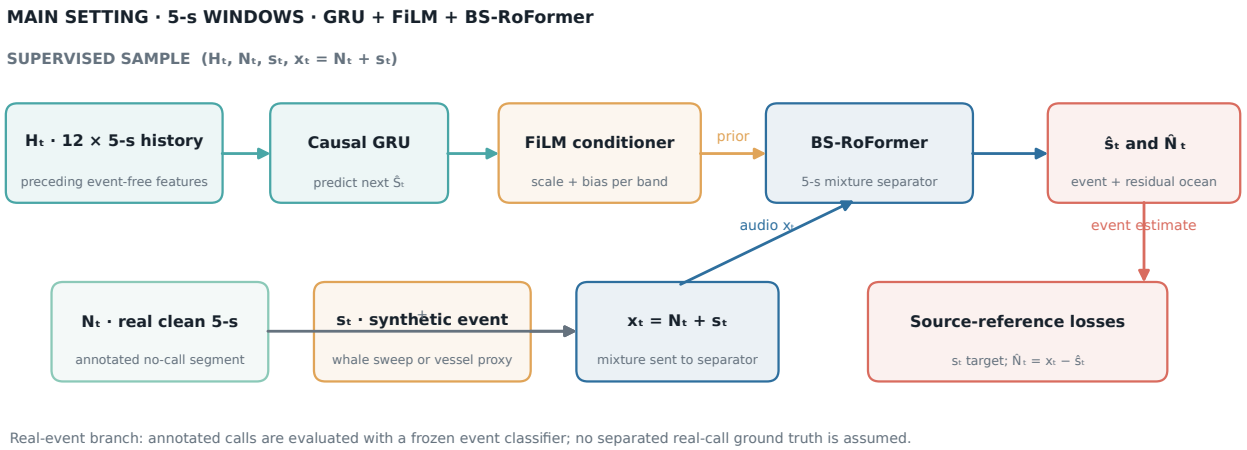


Fig. 1. The fixed main setting and its supervised data tuple. Event-masked history drives the GRU forecast; real annotated-clean background and a synthetic event form the current mixture. FiLM conditions the BS-RoFormer and the event estimate is supervised and the ocean estimate is its mixture-consistent residual.

C. Main data construction and secondary ablations

The main dataset uses day-disjoint 2-kHz DCLDE 2013 recordings and whale-call annotations [20], [21]. Across 168 h of WAV, excluding annotated calls and a 2-s guard on each side yields 146.4 h of annotation-clean 5-s windows. For each target, the 12 immediately preceding 5-s blocks from the same WAV form a 30-band history; a target is eligible only when all history blocks are outside annotated calls and their 2-s guards. This avoids temporal averaging over an hour and forbids histories from crossing WAV boundaries. This leaves 94.5 h eligible overall, split into 37.3/9.7/47.6 h for train/validation/test. For each current clean N_t, a 13-family synthetic event s_t is mixed to form $x_t = N_t + s_t$. The main split is by recording day, and the GRU receives 12 event-masked features from the immediately preceding 5-s windows.

Secondary ablations use NOAA–Navy SanctSound hourly TOL products [18]. MB03 deployments 01–02/03/04 are train/validation/test. Random-phase inversion creates controlled spectral surrogates, not recovered audio. Concatenation, cross-attention, KL/JS coverage, and the one-hour history-resolution comparison are confined to this domain. The separate 42-h SanctSound WAV strength sweep also lacks event annotations, so it is a useful controlled real-background ablation but is not counted as clean-background primary evidence.

Complete-reference tiers report SI-SDR and SI-SDRi using scale-invariant projection [22], [23]. All uncertainty intervals are seeded paired bootstraps, blocked by recording day for the main test. The event classifier is frozen before any real-event evaluation. Current test features are used only as targets for the synthetic full-reference task, never as GRU input; all histories end before t.

TABLE I
CONDITIONING-MODE ABLATION ON TOL-DERIVED SPECTRAL SURROGATES (DB).

Injection	Val. SI-SDRi	Test SI-SDRi	Paired Δ_0
None	4.864	4.059	0
Concatenation	5.343	4.432	+0.623
FiLM	**5.419**	**4.468**	**+0.692**
Cross-attention	5.128	4.261	+0.331

III. Experiments and Results

A. Conditioning-mode ablation on spectral surrogates

The primary architecture is fixed to FiLM. Table I compares it with the alternative interfaces on controlled 5-s spectral-surrogate examples; on 832 fixed-0-dB pairs (64 per family), FiLM improves 4.850 to 5.541 dB SI-SDRi. Band-wise scale and bias directly modulate existing band evidence. Concatenation must compress doubled features, while cross-attention must learn alignment from coarse predicted spectra under the same budget.

FiLM gains 0.869, 0.835, 0.919, 0.906, and 0.806 dB on the five whale proxy families. Vessel tonal is the hardest family: the conditioned absolute SI-SDRi is -8.295 dB, despite a 0.320-dB relative gain. Overall gain has weak correlation with JS ($r = -.057$) and entropy ($r = .122$), so increasing divergence broadens support but does not predict usefulness.

B. Primary fully supervised clean-background separation

The main FC-BSR receives twelve event-masked preceding 5-s features and a current clean 5-s DCLDE segment; the 13-family event is added only after the clean background is selected. The model therefore sees x_t and H_t, while the exact N_t and s_t are supervision. The training split contains 37.3 h of eligible clean current audio. The 12-step GRU forecasts the next 5-s band spectrum; validation MAE is 1.926 dB versus

TABLE II
PRIMARY HELD-OUT CLEAN-BACKGROUND + SYNTHETIC-EVENT TEST.

Model	SI-SDRi (dB)	Target SI-SDR (dB)	Background SI-SDR (dB)
BS-RoFormer	9.631	6.479	10.061
FC-BSR (GRU+FiLM)	9.497	6.345	10.155

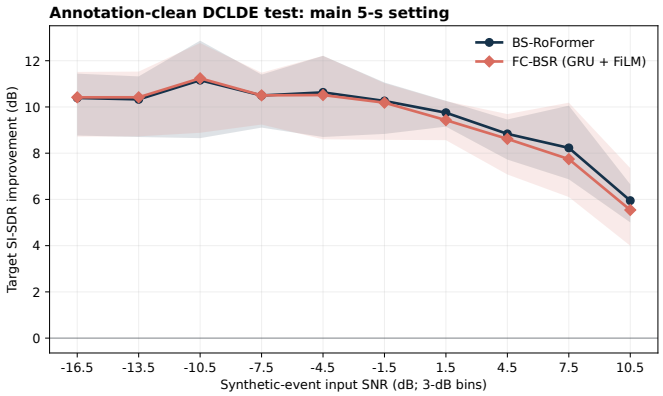


Fig. 2. Primary full-reference test on annotation-clean DCLDE backgrounds. Target SI-SDRi is shown against injected event strength for matched BS-RoFormer and FC-BSR; uncertainty is blocked by recording day.

TABLE III
ABLATION: PREDICTOR-RESOLUTION DIAGNOSTIC ON SANCTSOUND BACKGROUND.

Step	Predictor	Cond.	Base	Gain [95% CI]
5 s	GRU	29.753	28.527	1.226 [1.082,1.381]
30 s	Persist.	29.630	28.534	1.095 [0.935,1.265]
1 min	Persist.	29.661	28.533	1.128 [0.971,1.298]
10 min	Persist.	29.711	28.535	1.176 [1.036,1.326]

Cond./Base/Gain are target SI-SDRi in dB.

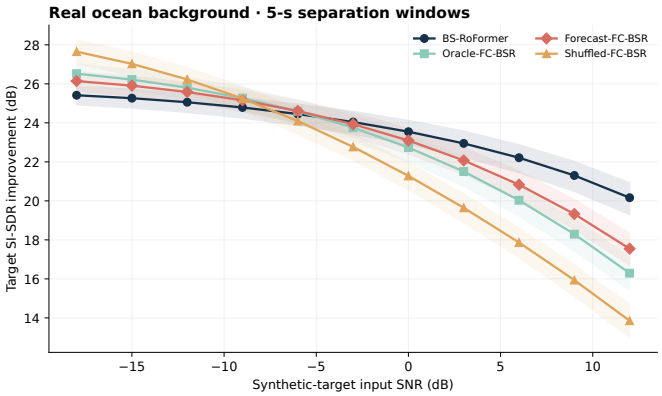


Fig. 3. Ablation on 5-s unlabeled SanctSound background: target-strength sweep. Lines show the four conditions; shaded bands are paired 95% intervals.

2.111 dB for persistence (8.8% lower). Table II reports the held-out full-reference comparison between the fixed main FC-BSR and its unconditioned BS-RoFormer control. The control shares data, initialization, optimizer, training duration, and checkpoint selection. Figure 2 stratifies this paired test by injected SNR while retaining the fixed main setting and held-out clean backgrounds. Across 4,096 paired examples, FC-BSR scores 9.497 dB SI-SDRi versus 9.631 dB for BS-RoFormer (paired difference -0.134 dB, day-block 95% CI $[-0.320, 0.125]$); the average difference is not distinguishable from zero.

C. Secondary ablations: time scale and target strength

The time-resolution diagnostic in Table III is an ablation on continuous SanctSound audio, not the main annotation-clean data. Only its 5-s predictor beats persistence on validation (MAE 1.366 versus 1.386 dB); the other scales select persistence. The separate 42-h strength curve also uses unlabeled current background. It shows that forecast conditioning is strength-dependent in that domain and must not be read as a result on annotation-clean current segments.

The harder fixed-5-s experiment compares BS-RoFormer, current-spectrum oracle, causal forecast, and time-shuffled condition on 1,152 paired test mixtures. Averaged across training SNRs, the baseline scores 23.470 dB and Forecast-FC-BSR 22.871 dB ($\Delta = -0.599$, CI $[-0.851, -0.346]$); even the oracle is lower. This exposes conditioning/capacity effects that the simpler diagnostic misses. Figure 3 gives the more informative strength sweep. Forecast conditioning helps weak targets by 0.728, 0.643, 0.525, 0.366, and 0.157 dB at $-18, -15, -12, -9, -6$ dB, respectively, then loses from -3 dB onward. The shuffled condition also helps the weakest targets but collapses as SNR rises; therefore weak-target gain alone cannot prove that temporal forecast accuracy, rather than regularization or extra capacity, is responsible.

D. Real events without separated ground truth

Following learned reference-free metrics [24], [25] and semantic audio embeddings [26], [27], we train, calibrate, and freeze a binary event classifier (test AUROC .9723, AP .9755, F1 .9275). Mixture consistency imposes $\hat{N} = x - \hat{s}$. We restore $\hat{s}$ at gains $\{-60, -30, -24, -18, -12, -6, 0\}$ dB and test whether event probability recovers monotonically. Identity and all-to-event controls reject trivial solutions; changes outside annotated intervals audit preservation.

The same frozen 12-step event-masked forecaster used by the main setting supplies the causal history features. Because recent call activity is masked, 103 real test calls retain a full eligible 60-s history. Table IV shows the untouched-call results: FC-BSR trends toward lower residual event probability and off-event change, while event-stem probability is nearly unchanged. The day-block intervals include zero for residual probability $\Delta = -.0731$ ($[-.2090, .0294]$), event-stem probability $\Delta = +.0022$ ($[-.0180, .0304]$), and monotonicity $\Delta = +.0178$ ($[-.0333, .0429]$). Off-event relative error decreases by .0248 ($[-.0456, -.0018]$). This small semantic set is suggestive but not decisive; separated-stem ground truth is still unavailable.

IV. DISCUSSION AND CONCLUSION

The evidence supports a narrower claim than the initial hypothesis. On the primary annotation-clean test, FC-BSR does

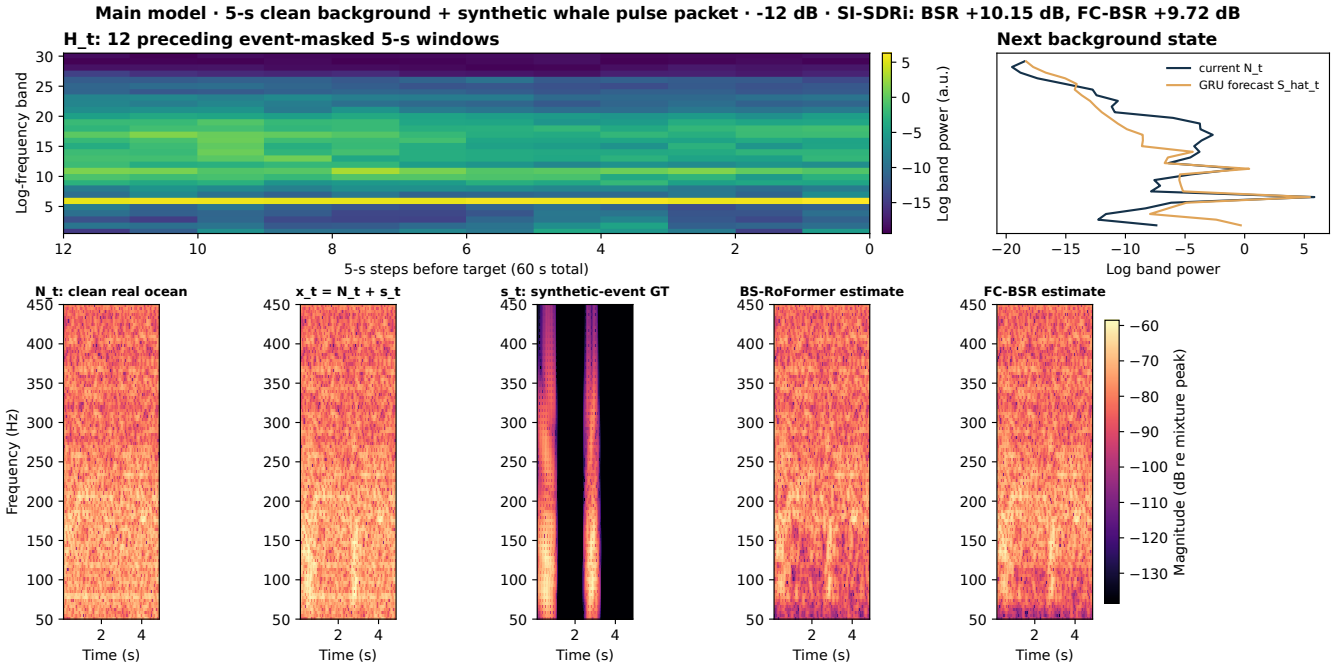


Fig. 4. Median-selected −12 dB synthetic whale pulse packet in a current annotation-clean 5-s DCLDE recording. All spectrograms share one magnitude scale referenced to the mixture peak; the aggregate strength results are in Fig. 2.

TABLE IV
REFERENCE-FREE MATCHED COMPARISON ON 103 REAL CALLS WITH ELIGIBLE HISTORIES.

Method	$p(x)$	$p(\hat{N})\downarrow$	$p(\hat{s})\uparrow$	Mono.↑	Off-event↓
BS-RoFormer	.909	.786	.900	.937	.230
FC-BSR	.909	.713	.902	.955	.206

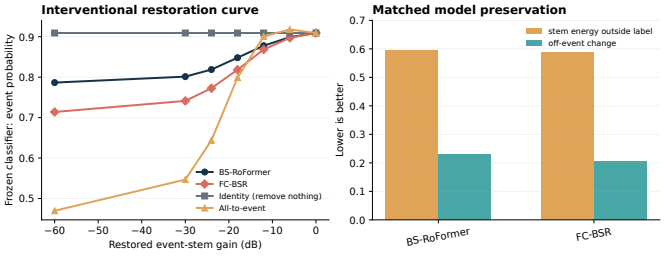


Fig. 5. Reference-free restoration curves and preservation audit. Both models perform nontrivial semantic separation, but FC-BSR is not uniformly better on real calls.

not improve mean SI-SDRi over the matched BS-RoFormer; the 95% interval spans zero. FiLM is the best conditioning interface on controlled stems. In the secondary unannotated-WAV strength ablation, the prior helps weak targets, but that result does not establish a gain on the annotation-clean main setting. It is not universally beneficial: the gain reverses for stronger targets, an oracle condition does not rescue the average, and no decisive forecast advantage appears on untouched whale calls. Persistence also wins three of four predictor resolutions. These negative controls are central: they separate the existence of useful background information from a claim that the present architecture always exploits it.

The local frontend is usable for inspection: on a CPU smoke test, one 5-s chunk ran at RTF 0.014 after history features were prepared; the interface displays forecasts and permits playback and download of both estimates. This does not establish field deployment. Future work should condition only under calibrated uncertainty, test multiple sites and seasons, and obtain isolated or stronger partial references for real events. Classifier-based restoration is evidence of semantic transfer, not a replacement for waveform fidelity.

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